

SIMATS ENGINEERING ASSESSMENT TOOLS

COURSE NAME: Theory of Computation **COURSE CODE:** CSA1304

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:100

Duration : 1 Hour

CLASS TEST

Code of Conduct:

I Yuvann Nithish R (192411267) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Yuvann Nithish R

SIMATS ENGINEERING ASSESSMENT TOOLS

1. Design a PDA to accept strings with 'n' occurrences of 'a' followed by 'n' occurrences of 'b', where 'n' is a non-negative integer.

i) $L = \{a^n b^n\}$

2. Convert the given context-free grammar (CFG) into a Pushdown Automaton (PDA) that recognizes the same language.

$E \rightarrow E+E \mid E^*E \mid (E) \mid I$

$I \rightarrow Ia \mid Ib \mid IO \mid I1 \mid a \mid b$

3. Design a TM to accept the strings with 'n' occurrences of 'a' followed by 'n' occurrences of 'b', where 'n' is a non-negative integer

i) $L = \{0^n 1^n \mid n \geq 1\}$

4. Design a Pushdown Automata for the language representing a's followed by b's and the number of b's must be twice that of a's.

$$L = \{a^n b^{2n} \mid n \geq 1\}$$

5. Construct a Turing Machine to recognize the language. Write the logic used for the design and the formal definition

$$L = \{a^n b^{2n} \mid n \geq 1\}$$

Formal Languages and Automata Theory — Solutions

1. Design a PDA for $(L = \{a^n b^n \mid n \geq 0\})$

Idea

The PDA must compare the number of a s with the number of b s.

For every a , push one A onto the stack.

When b starts, pop one A for every b .

If all A s are removed exactly when the input ends, accept.

Since $(n \geq 0)$, the empty string (ϵ) is also accepted.

PDA Definition

[
 $M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$
]

where:

[
 $Q = \{q_0, q_1, q_f\}$
]

[
 $\Sigma = \{a, b\}$
]

[
 $\Gamma = \{A, Z_0\}$
]

Initial state:

[
 q_0
]

Initial stack symbol:

$$\begin{bmatrix} Z_0 \end{bmatrix}$$

Final states:

$$\begin{bmatrix} F = \{q_f\} \end{bmatrix}$$

Transitions

1.

For the first a :

2.

$$\begin{bmatrix} \delta(q_0, a, Z_0) = \{(q_0, AZ_0)\} \end{bmatrix}$$

1.

For subsequent a s:

2.

$$\begin{bmatrix} \delta(q_0, a, A) = \{(q_0, AA)\} \end{bmatrix}$$

1.

On reading the first b :

2.

$$\begin{bmatrix} \delta(q_0, b, A) = \{(q_1, \epsilon)\} \end{bmatrix}$$

For remaining b s

$$\begin{bmatrix} \delta(q_1, b, A) = \{(q_1, \epsilon)\} \end{bmatrix}$$

When all A s are removed:

$$[\delta(q_1, \epsilon, Z_0) = \{(q_f, Z_0)\}]$$

For (n=0):

$$[\delta(q_0, \epsilon, Z_0) = \{(q_f, Z_0)\}]$$

Example

For input:

$$[aaabbb]$$

Stack operation:

$$\begin{aligned} a &\rightarrow AZ0 \\ a &\rightarrow AAZ0 \\ a &\rightarrow AAAZ0 \\ b &\rightarrow AAZ0 \\ b &\rightarrow AZ0 \\ b &\rightarrow Z0 \end{aligned}$$

Therefore the PDA reaches (q_f).

$$[\boxed{\text{aaabbb is accepted}}]$$

Hence,

$$[\boxed{L(P) = \{a^n b^n \mid n \geq 0\}}]$$

2. Convert the CFG into a PDA

Given CFG:

$$[E \rightarrow E + E \mid E * E \mid (E) \mid I]$$

$$[\rightarrow I_a \mid I_b \mid I_0 \mid I_1 \mid a \mid b]$$

Construction

For converting a CFG to PDA:

1.

Push the start variable E onto the stack.

2.

3.

Replace a variable using an (ϵ)-transition according to its production.

4.

5.

If the stack top is a terminal matching the current input symbol, read the symbol and pop it.

6.

7.

Accept when the entire input has been consumed.

8.

PDA

Let

$$[P = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)]$$

with:

$$[Q = \{q_0, q, q_f\}]$$

The stack initially contains:

$$[EZ_0]$$

Production transitions

For:

[
 $E \rightarrow E + E$
]

[
 $\delta(q, \epsilon, E) = (q, E + E)$
]

For:

[
 $E \rightarrow E * E$
]

[
 $\delta(q, \epsilon, E) = (q, E * E)$
]

For:

[
 $E \rightarrow (E)$
]

[
 $\delta(q, \epsilon, E) = (q, (E))$
]

For:

[
 $E \rightarrow I$
]

[
 $\delta(q, \epsilon, I) = (q, I)$
]

For (I):

[
 $\delta(q, \epsilon, I) = (q, I)$
]

$$[\delta(q, \epsilon, I) = (q, I_b)]$$

$$[\delta(q, \epsilon, I) = (q, I_0)]$$

$$[\delta(q, \epsilon, I) = (q, I_1)]$$

$$[\delta(q, \epsilon, I) = (q, a)]$$

$$[\delta(q, \epsilon, I) = (q, b)]$$

Terminal matching

For every terminal:

$$[\delta(q, a, a) = (q, \epsilon)]$$

$$[\delta(q, b, b) = (q, \epsilon)]$$

$$[\delta(q, 0, 0) = (q, \epsilon)]$$

$$[\delta(q, 1, 1) = (q, \epsilon)]$$

$$[\delta(q, +, +) = (q, \epsilon)]$$

$$[\delta(q, \epsilon) = (q, \epsilon)]$$

Similarly for (and) .

When the input is completely consumed and only (Z_0) remains:

```
[  
\delta(q,\epsilon,Z_0)=(q_f,Z_0)  
]
```

Example

For the string:

```
[  
a+b  
]
```

A possible derivation is:

```
[  
E \rightarrow E+E  
]
```

```
[  
 \rightarrow I+E  
]
```

```
[  
 \rightarrow a+E  
]
```

```
[  
 \rightarrow a+I  
]
```

```
[  
 \rightarrow a+b  
]
```

Therefore:

```
[  
\boxed{a+b \text{ is accepted}}  
]
```

3. Design a TM for $(L = \{0^n 1^n \mid n \geq 1\})$

Logic

The TM uses two marker symbols:

-

$x \rightarrow \text{matched } 0$

-

-

$y \rightarrow \text{matched } 1$

-

Algorithm

- 1.

Start from the left.

- 2.

- 3.

Find the first unmarked 0.

- 4.

- 5.

Replace it with x.

- 6.

- 7.

Move right until an unmarked 1 is found.

- 8.

- 9.

Replace that 1 with y.

- 10.

- 11.

Move back to the left.

- 12.

- 13.

Repeat the process.

14.

15.

If all 0s are marked and all 1s are also marked, accept.

16.

17.

If a matching 1 is missing, reject.

18.

19.

If a 1 occurs before an unmatched 0, reject.

20.

Important transitions

[
 $\delta(q_0, 0) = (q_1, X, R)$
]

[
 $\delta(q_1, 0) = (q_1, 0, R)$
]

[
 $\delta(q_1, Y) = (q_1, Y, R)$
]

[
 $\delta(q_1, 1) = (q_2, Y, L)$
]

In (q_2) , move left:

[
 $\delta(q_2, 0) = (q_2, 0, L)$
]

[
 $\delta(q_2, Y) = (q_2, Y, L)$
]

When x is found:

[
 $\delta(q_2, X) = (q_0, X, R)$
]

After all 0s have been matched, scan the remaining 1s:

[
 $\delta(q_4, Y) = (q_4, Y, R)$
]

On blank:

[
 $\delta(q_4, B) = (q_f, B, S)$
]

Example

Input:

[
 0011
]

Working:

0011
 X011
 X0Y1
 XXY1
 XXYY

All symbols are matched.

[
 $\boxed{0011 \text{ is accepted}}$
]

Similarly:

[
 000111
]

is accepted.



4. Design a PDA for $(L = \{a^n b^{2n} \mid n \geq 1\})$

Idea

The number of b s must be **twice** the number of a s.

Therefore:

•

For every a , push **two** A s.

•

•

For every b , pop **one** A .

•

•

Accept when all A s are removed exactly at the end.

•

PDA Definition

[
M=(Q,\Sigma,\Gamma,\delta,q_0,Z_0,F)
]

where:

[
Q={q_0,q_1,q_f}
]

[
\Sigma={a,b}
]

[
\Gamma={A,Z_0}
]

$$[$$

$$F=\{q_f\}$$

$$]$$

Transitions

For the first a :

$$[$$

$$\delta(q_0, a, Z_0) = \{(q_0, AAZ_0)\}$$

$$]$$

For every additional a :

$$[$$

$$\delta(q_0, a, A) = \{(q_0, AAA)\}$$

$$]$$

Notice that replacing one existing A by AAA increases the number of A s by two.

When b starts:

$$[$$

$$\delta(q_0, b, A) = \{(q_1, \epsilon)\}$$

$$]$$

For subsequent b s:

$$[$$

$$\delta(q_1, b, A) = \{(q_1, \epsilon)\}$$

$$]$$

Finally:

$$[$$

$$\delta(q_1, \epsilon, Z_0) = \{(q_f, Z_0)\}$$

$$]$$

Example

Input:

$$[$$

$$aabbbb$$

$$]$$

After first a :

$$[Z_0 \rightarrow AAZ_0]$$

After second a:

$$[AAZ_0 \rightarrow AAAZ_0]$$

Now process four bs:

$$\begin{aligned} b &\rightarrow AAAZ_0 \\ b &\rightarrow AAZ_0 \\ b &\rightarrow AZ_0 \\ b &\rightarrow Z_0 \end{aligned}$$

Therefore:

$$[\boxed{\text{aabb}} \text{ is accepted}]$$

Hence:

$$[\boxed{L(P) = \{a^n b^{2n} \mid n \geq 1\}}]$$

5. Construct a TM for $(L = \{a^n b^n c^n \mid n \geq 1\})$

The expression in the original question was corrupted/unreadable. I am using the standard interpretation $(L = \{a^n b^n c^n \mid n \geq 1\})$.

Logic

The TM uses three marker symbols:

-
- $x \rightarrow \text{matched } a$
-

•

$Y \rightarrow \text{matched } b$

•

•

$Z \rightarrow \text{matched } c$

•

For every one a , the machine must find exactly one b and one c .

Algorithm

1.

Find the leftmost unmarked a .

2.

3.

Replace it with x .

4.

5.

Move right through the a s and marked symbols.

6.

7.

Find the first unmarked b .

8.

9.

Replace it with y .

10.

11.

Continue moving right.

12.

13.

Find the first unmarked c .

14.

15.

Replace it with z .

16.

17.

Move back to the left end.

18.

19.

Repeat.

20.

21.

When no unmarked a remains, verify that only y s and z s remain.

22.

23.

If everything is correct, accept.

24.

25.

Otherwise, reject.

26.

Formal Definition

[
 $T = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$
]

where:

[
 $Q = \{q_0, q_1, q_2, q_3, q_4, q_{\text{accept}}, q_{\text{reject}}\}$
]

[
 $\Sigma = \{a, b, c\}$
]

[
\\Gamma={a,b,c,X,Y,Z,B}
]

Initial state:

[
q_0
]

Blank:

[
B
]

Final state:

[
F={q_{accept}}
]

Main transitions

Find an a:

[
\\delta(q_0,a)=(q_1,X,R)
]

Search for b:

[
\\delta(q_1,a)=(q_1,a,R)
]

[
\\delta(q_1,Y)=(q_1,Y,R)
]

[
\\delta(q_1,b)=(q_2,Y,R)
]

Search for c:

[
\\delta(q_2,b)=(q_2,b,R)
]

[
 \delta(q_2,Z)=(q_2,Z,R)
]

[
 \delta(q_2,c)=(q_3,Z,L)
]

Move back toward the beginning:

[
 \delta(q_3,a)=(q_3,a,L)
]

[
 \delta(q_3,b)=(q_3,b,L)
]

[
 \delta(q_3,Y)=(q_3,Y,L)
]

[
 \delta(q_3,Z)=(q_3,Z,L)
]

When x is reached:

[
 \delta(q_3,X)=(q_0,X,R)
]

Example

Input:

[
 aabbcc
]

Working:

aabbcc
Xabbcc
XaYbcc
XaYbZc
XXYbZc
XXYYZc
XXYYZZ

All three groups have the same number of symbols.

Therefore:

[
\boxed{aabbcc\text{ is accepted}}
]

For example:

[
abc,\quad aabbcc,\quad aaabbbccc
]

are accepted.

But:

[
aabbbccc
]

is rejected because the number of b s is different from the number of a s.

Final Answers at a Glance

Question	Language	Main Technique
1	$(a^n b^n)$	PDA: push 1, pop 1
2	Given CFG	CFG $\rightarrow$ PDA using ϵ -transitions
3	$(0^n 1^n)$	TM: mark $0 \rightarrow X$, $1 \rightarrow Y$
4	$(a^n b^{2n})$	PDA: push 2, pop 1
5	$(a^n b^n c^n)$	TM: mark $a \rightarrow X$, $b \rightarrow Y$, $c \rightarrow Z$

SIMATS ENGINEERING ASSESSMENT TOOLS

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficiency (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
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Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand